

NUS QF3101: Investment Instrument and Risk Management

April 1, 2023

Computing Assignment

Instructor: Tan Hwee Huat

Student: Wira Azmoon Ahmad A0149286R

1 Three-month SOFR Futures and Deferred IRS

1.a Data Collection and Preparation

THREE-MONTH SOFR FUTURES - QUOTES

AUTO-REFRESH 10 SECS

Last Updated On Mar-2023 03:22 AM CDT

Market Data is delivered by JPMarkets US (in hours)

VENUE: GLOBEX

MONTHS

OPTIONS

CHART

LAST

CHANGE

PRICE OFFER

OPEN

HIGH

LOW

VOLUME

UPDATED

JUN 2027

SO27F

☑

⬇

95.78

-0.85 (-0.85%)

95.83

95.745

95.885

95.74

5,515

01:02:06 CT
15 Mar 2023

JUN 2027

SO27F

☑

⬇

95.78

-0.845 (-0.85%)

95.825

95.755

95.885

95.74

5,319

01:28:58 CT
15 Mar 2023

SEP 2027

SO27F

☑

⬇

95.74

-0.84 (-0.84%)

95.78

95.705

95.77

95.70

6,773

01:29:57 CT
15 Mar 2023

DEC 2027

SO27F

☑

⬇

95.785

-0.84 (-0.84%)

95.785

95.67

95.73

95.66

5,234

01:21:08 CT
15 Mar 2023

MAR 2028

SO28F

☑

⬇

95.785

-0.84 (-0.84%)

95.785

95.685

95.725

95.66

3,385

01:29:57 CT
15 Mar 2023

MAY 2028

SO28F

☑

⬇

95.665

-0.875 (-0.88%)

95.74

95.67

95.67

95.665

34

00:55:00 CT
15 Mar 2023

SEP 2028

SO28F

☑

⬇

-

-

95.715

-

-

-

9

00:46:55 CT
15 Mar 2023

DEC 2028

SO28F

☑

⬇

-

-

95.67

-

-

-

7

00:46:55 CT
15 Mar 2023

MAR 2029

SO29F

☑

⬇

-

-

95.645

-

-

-

0

22:13:12 CT
14 Mar 2023

JUN 2029

SO29F

☑

⬇

-

-

95.615

-

-

-

0

22:13:12 CT
14 Mar 2023

SEP 2029

SO29F

☑

⬇

-

-

95.59

-

-

-

0

22:13:12 CT
14 Mar 2023

DEC 2029

SO29F

☑

⬇

-

-

95.565

-

-

-

0

19:26:47 CT
14 Mar 2023

MAR 2030

SO30F

☑

⬇

-

-

95.55

-

-

-

0

19:27:00 CT
14 Mar 2023

JUN 2030

SO30F

☑

⬇

-

-

95.53

-

-

-

0

22:13:12 CT
14 Mar 2023

SEP 2030

SO30F

☑

⬇

-

-

95.52

-

-

-

0

22:13:12 CT
14 Mar 2023

DEC 2030

SO30F

☑

⬇

-

-

95.51

-

-

-

0

22:13:12 CT
14 Mar 2023

MAR 2031

SO31F

☑

⬇

-

-

95.515

-

-

-

0

19:28:05 CT
14 Mar 2023

JUN 2031

SO31F

☑

⬇

-

-

95.50

-

-

-

0

19:27:12 CT
14 Mar 2023

SEP 2031

SO31F

☑

⬇

-

-

95.485

-

-

-

0

22:13:12 CT
14 Mar 2023

DEC 2031

SO31F

☑

⬇

-

-

95.475

-

-

-

0

19:27:19 CT
14 Mar 2023

MAR 2032

SO32F

☑

⬇

-

-

95.465

-

-

-

0

19:27:43 CT
14 Mar 2023

JUN 2032

SO32F

☑

⬇

-

-

95.455

-

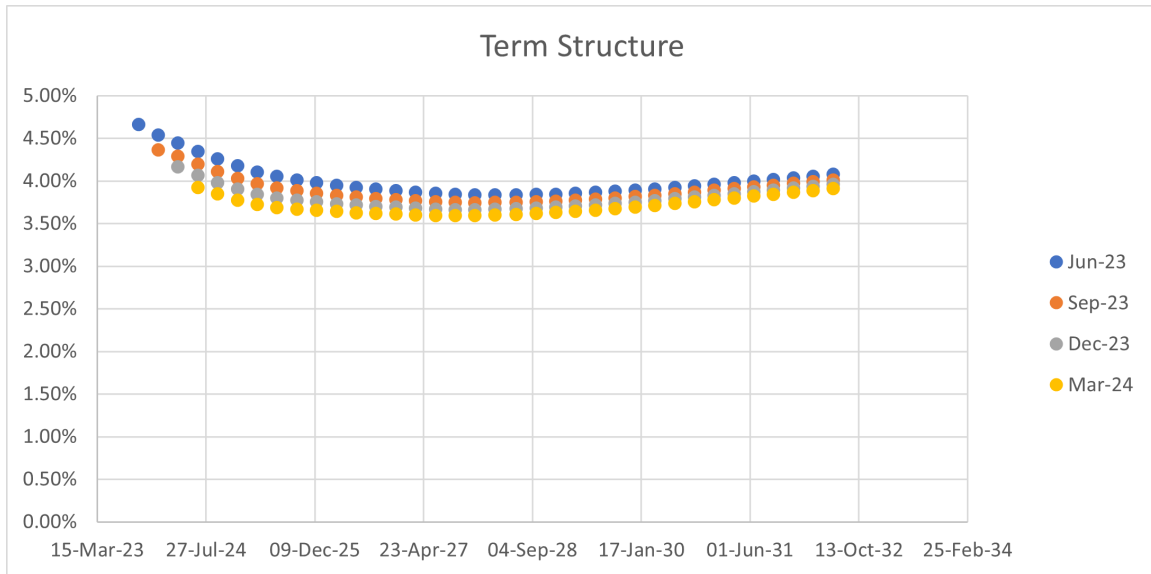
-

-

0

19:26:47 CT
14 Mar 2023

1.b Implied Forward SOFR Term Structure



The term structure is downward sloping for at least the next 4 years, which signifies that the economy is in, or about to enter, a recessive period. There is a slight uptrend afterwards till the end of the decade, signifying a predicted expansionary period. The term structure is also generally lower for later reference dates, but match up to similar values further into the future. This signifies the uncertainty between the now and the future.

1.c Swap Rate for Deferred Interest Rate Swap

K	1	2	3	4	5	6	7	8
Jun-23	4.36%	3.99%	3.80%	3.67%	3.61%	3.57%	3.56%	3.56%
Sep-23	4.12%	3.82%	3.68%	3.57%	3.53%	3.51%	3.51%	3.51%
Dec-23	3.91%	3.70%	3.58%	3.50%	3.47%	3.47%	3.47%	3.48%
Mar-24	3.73%	3.59%	3.49%	3.44%	3.43%	3.43%	3.44%	3.46%

Similar to the term structure, the swap rate drops for longer swap periods. It is also lower for further deferred start periods compared to earlier.

2 Efficient Equity Portfolios

2.a Data collection and preparation.

Average rate of return

$$r = \begin{bmatrix} 0.5583\% & 0.1963\% & 0.1510\% & 0.2355\% & -0.0618\% \end{bmatrix}^T$$

Covariance matrix

$$C = \begin{bmatrix} 0.00176 & 0.00064 & 0.00061 & 0.00099 & 0.00074 \\ 0.00064 & 0.00094 & 0.00070 & 0.00111 & 0.00080 \\ 0.00061 & 0.00070 & 0.00179 & 0.00100 & 0.00075 \\ 0.00099 & 0.00111 & 0.00100 & 0.00392 & 0.00179 \\ 0.00074 & 0.00080 & 0.00075 & 0.00179 & 0.00297 \end{bmatrix}$$

On average, AAPL performs best, while HAS performs worst. The covariance matrix is entirely positive, so it indicates that all 5 stocks tend to move in the same direction.

2.b Minimum Variance Point Portfolio And An Efficient Portfolio.

Minimum Variance portfolio with weight vector

$$w_G = \begin{bmatrix} 0.19281 & 0.69828 & 0.15050 & -0.10442 & 0.06284 \end{bmatrix}^T$$

with expected rate of return

$$r_G = 0.2390\%$$

and variance 0.000818.

An efficient portfolio with weight vector

$$w_m = \begin{bmatrix} 1.159769 & 0.311124 & -0.061948 & 0.049304 & -0.458249 \end{bmatrix}^T$$

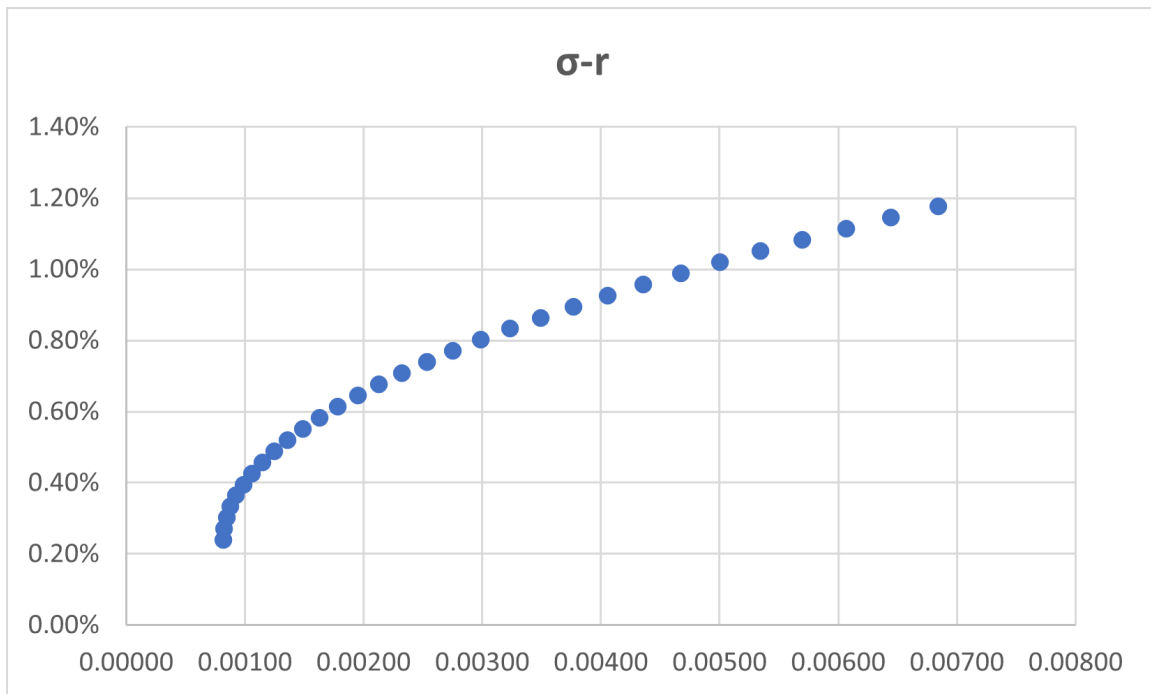
with expected rate of return

$$r_m = 0.7392\%$$

and variance 0.002531.

It is clear that any portfolio other than the minimum variance portfolio will have a higher variance.

2.c Efficient Portfolios



Using the Two-Fund Theorem, we can choose any portfolio with weights

$$w = \alpha w_m + (1 - \alpha)w_G$$

where $\alpha > 0$ to have an efficient portfolio with return $r > r_G$. These portfolios are on the efficient frontier, visible on the $\sigma - \bar{r}$ graph above.